

Production Networks & Monetary Policy

A Complete Guide: From Scratch to Rubbo (2024) to SOE Extension

Companion document to `production_network_guide.m`

Working Notes

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1 What the Matlab file contains

The file `production_network_guide.m` is a single runnable Matlab script (run section by section with `Ctrl+Enter`) that covers three topics:

1. **Part 1:** Production networks from scratch
2. **Part 2:** Rubbo (2024) formula $\rightarrow$ code, line by line
3. **Part 3:** Small open economy extension (your paper)

Every formula is explained in plain English *before* the code appears. The file also contains three utility functions at the end: `soe_parameters()`, `optimal_fx_intervention()`, and `plot_welfare_decomposition`

2 Part 1 — Production Networks from Scratch

2.1 The IO matrix Ω

The economy is a web of industries. The input-output (IO) matrix Ω encodes who buys from whom:

$$\Omega_{ij} = \text{fraction of sector } i\text{'s total costs spent on sector } j\text{'s goods}$$

Each row of Ω is sector i 's “shopping basket” for intermediate inputs. The rest of sector i 's costs go to workers: the **labor share** is $\alpha_i = 1 - \sum_j \Omega_{ij}$.

Cost identity: for every sector i , $\alpha_i + \sum_j \Omega_{ij} = 1$ (all revenue goes to either labor or intermediate inputs).

2.2 The Leontief inverse

If Energy prices rise by 1%, Manufacturing's costs go up not just by $\Omega_{\text{Manuf}, \text{Energy}}$ (direct effect), but also through indirect chains: Manufacturing buys Services which uses Energy; those Services firms raise their prices, which feeds back into Manufacturing's costs, and so on.

The **Leontief inverse** $(I - \Omega)^{-1}$ captures *all* rounds:

$$(I - \Omega)^{-1} = I + \Omega + \Omega^2 + \Omega^3 + \dots$$

where Ω_{ij}^k is the k -step indirect input flow from sector j to sector i .

Listing 1: Matlab: Leontief inverse

```
L = inv(eye(n) - Omega);           % exact
L_approx = eye(n) + Omega + Omega^2 + Omega^3 + ...; % geometric series
```

2.3 Domar weights

The **Domar weight** of sector i is its total sales as a fraction of GDP, counting both direct sales to consumers and indirect sales through other producers:

$$\lambda^T = \beta^T (I - \Omega)^{-1}$$

Key property: $\sum_i \lambda_i > 1$ always, because Domar weights measure *gross* output, not value added.

Listing 2: Matlab: Domar weights

```
lambda = beta' * inv(eye(n) - Omega);    % 1 x N row vector
```

Why Domar weights matter for MP: The output gap equals minus the Domar-weighted average markup (Rubbo Lemma 3): $(\gamma + \varphi)\tilde{y}_t = -\sum_i \lambda_i \mu_{it}$. An upstream sector (large λ_i) causes more macroeconomic damage when it prices above marginal cost than a downstream sector of the same direct size.

2.4 TFP propagation through the network (Lemma 1)

Natural output is Domar-weighted TFP:

$$y_{\text{nat}} = \frac{1 + \varphi}{\gamma + \varphi} \lambda^T \log \mathbf{A}$$

A 1% TFP gain in sector i raises natural output by $\frac{1+\varphi}{\gamma+\varphi} \lambda_i$, which is *larger* than $\frac{1+\varphi}{\gamma+\varphi} \beta_i$ because $\lambda_i > \beta_i$: the network amplifies the effect of upstream sectors.

3 Part 2 — Rubbo (2024): Formula to Matlab, Line by Line

3.1 Effective Calvo parameters $\hat{\delta}_i$

In the Calvo model, sector i resets its price with probability δ_i per period. The log-linearized model uses a slightly adjusted parameter:

$$\hat{\delta}_i = \frac{\delta_i(1 - \rho(1 - \delta_i))}{1 - \rho\delta_i(1 - \delta_i)}$$

For $\rho \approx 1$ this is very close to δ_i . In matrix form, $\Delta = \text{diag}(\hat{\delta}_1, \dots, \hat{\delta}_N)$.

Listing 3: Matlab: effective Calvo parameters

```
hat_delta = delta .* (1 - rho*(1-delta)) ./ (1 - rho*delta.*(1-delta));
Delta = diag(hat_delta);
```

3.2 The rigidity-adjusted Leontief $(I - \Omega\Delta)^{-1}$

This is the **central object** in Rubbo's Phillips curve. Compare:

$$\text{Standard Leontief: } (I - \Omega)^{-1} = I + \Omega + \Omega^2 + \dots$$

$$\text{Rigidity-adjusted: } (I - \Omega\Delta)^{-1} = I + \Omega\Delta + (\Omega\Delta)^2 + \dots$$

Each round of the supply chain is now weighted by Δ : a sticky-price sector (small $\hat{\delta}_j$) absorbs cost shocks into its markup rather than passing them on, so the cost ripple through the network is attenuated.

The key intermediate matrix appearing everywhere in Rubbo's formulas:

$$\tilde{A} \equiv \Delta(I - \Omega\Delta)^{-1}$$

Listing 4: Matlab: rigidity-adjusted Leontief and key intermediate

```
L_rigid = inv(eye(n) - Omega*Delta);
A_tilde = Delta * L_rigid;
kappa   = 1 - beta'*A_tilde*alpha;    % GE normalization scalar
```

3.3 Phillips curve slope $\mathcal{B}$ (Proposition 2)

The sector-level Phillips curves are (Proposition 2):

$$\pi_t = \rho(I - \mathcal{V}) \mathbb{E}\pi_{t+1} + \mathcal{B}(\gamma + \varphi) \tilde{y}_t - \mathcal{V}\chi_t$$

The Phillips curve slope $\mathcal{B}$ (vector):

$$\mathcal{B} = \frac{\tilde{A}\alpha}{\kappa} \quad \text{where } \kappa = 1 - \beta^T \tilde{A}\alpha$$

$\mathcal{B}_i$ = how much sector i 's inflation rises per unit output gap. Larger when: (a) prices are flexible (large $\hat{\delta}_i$, captured in Δ); (b) labor costs are important (large α_i); (c) the sector's suppliers also have (a) and (b) (captured by $(I - \Omega\Delta)^{-1}$).

Listing 5: Matlab: Phillips curve slope (matches parameters.d.m)

```
b = (gamma+phi) * A_tilde * alpha / kappa;    % N x 1 vector
```

3.4 Cost-push matrix $\mathbf{v}$ (Proposition 2)

The cost-push matrix $\mathcal{V}$ (called $\mathbf{v}$ in Rubbo's code) multiplies the cost-push state $\mathbf{x}_t = (I - \Omega)^{-1} \log \mathbf{A}_t - \mathbf{p}_{t-1}$:

$$\mathcal{V} = \tilde{A} \left(\boldsymbol{\alpha} \frac{\lambda^T - \beta^T \tilde{A}}{\kappa} - I \right)$$

Key property: rows of $\mathcal{V}(I - \Omega)$ sum to zero. This means *uniform* cost shocks (all sectors equally productive) generate zero cost-push inflation — only *relative* price distortions matter.

Listing 6: Matlab: cost-push matrix (matches parameters_d.m)

```
v = A_tilde * (alpha*(lambda - beta'*A_tilde)/kappa - eye(n));
% Verification: v*(I-Omega)*ones should be zero
disp(v*(eye(n)-Omega)*ones(n,1))
```

3.5 Divine Coincidence index weights (Proposition 1)

The DC index $\pi_t^{DC} = (\mathbf{w}^{DC})^T \boldsymbol{\pi}_t$ is the **unique** inflation measure satisfying a cost-push-free Phillips curve:

$$\pi_t^{DC} = \rho \mathbb{E} \pi_{t+1}^{DC} + \frac{\gamma + \varphi}{\Lambda^{DC}} \tilde{y}_t$$

DC weights (unnormalized): $w_i^{DC} = \lambda_i \frac{1 - \hat{\delta}_i}{\hat{\delta}_i}$

Sectors get higher DC weight when: (i) λ_i is large (big network footprint); (ii) $\hat{\delta}_i$ is small (sticky prices $\Rightarrow$ markups respond more to inflation).

Listing 7: Matlab: DC index weights (matches parameters_d.m)

```
llambda      = lambda * (eye(n)-Delta) * inv(Delta);    % 1 x N, unnormalized
dc_weights   = llambda / sum(llambda);                  % normalize
phi_DC       = (gamma+phi) / sum(llambda);              % DC Phillips slope
```

3.6 System matrices $\mathbf{MM}$ and $\mathbf{QZ}$ solution

After assembling all equations, the model becomes a linear RE system $A \mathbb{E}[x_{t+1}] = B x_t$. Rubbo builds the system matrix using:

$$\mathbf{MM} = I + \Delta \mathbf{v} (I - \Omega)$$

and solves via the $\mathbf{QZ}$ (generalized Schur) decomposition. The solution:

$$(\text{jump variables})_t = \mathbf{g} \mathbf{x} \cdot \text{state}_t, \quad \text{state}_{t+1} = \mathbf{h} \mathbf{x} \cdot \text{state}_t$$

Listing 8: Matlab: system matrix and QZ solve

```
MM = eye(n) + Delta*v*(eye(n)-Omega);
[S,T,Q,Z] = qz(A,B);           % QZ decomposition
slt = abs(diag(T)) < abs(diag(S)); % stable eigenvalues
[S,T,Q,Z] = ordqz(S,T,Q,Z,slt); % reorder: stable first
gx = real(Z21 * Z11^(-1));      % jump variables = gx * state
hx = real(Z11 * (S11\T11) * Z11^(-1)); % state transition
```

4 Part 3 — Small Open Economy Extension

4.1 Modified marginal cost

Domestic firms now also use imported inputs with cost share ω_{Fi} . Import costs = world price p^* + log exchange rate e (depreciation $\Rightarrow e$ rises $\Rightarrow$ imports more expensive). Modified marginal cost:

$$mc_t = \alpha w_t + \Omega \mathbf{p}_t + \omega_F(p_t^* + e_t) - \log \mathbf{A}_t$$

Same model, one new term. Everything else follows identically.

4.2 Exchange rate pass-through vector Γ

Going through Steps 1–7 of Proposition 2 with the modified mc gives:

$$\pi_t = \rho(I - \mathcal{V}) \mathbb{E} \pi_{t+1} + \mathcal{B}(\gamma + \varphi) \tilde{y}_t - \mathcal{V} \chi_t + \Gamma \Delta e_t$$

where $\Delta e_t = e_t - e_{t-1}$ (depreciation) and:

$$\Gamma = \frac{\tilde{A} \omega_F}{\kappa}$$

Same formula as $\mathcal{B}$, but ω_F replaces α . This is the key: the derivation is identical because $\omega_F(p^* + e)$ enters mc in exactly the same position as αw — it's just a different cost component.

Listing 9: Matlab: ER pass-through vector (SOE extension)

```
Gamma = A_tilde * omega_F / kappa;    % N x 1
% Interpretation: Gamma_i = sector i's inflation per 1% depreciation
% Gamma_i > omega_F_i because indirect import exposure through supply chains
```

Network amplification of ER pass-through: $\Gamma_i > \omega_{Fi}$ because the rigidity-adjusted Leontief $(I - \Omega \Delta)^{-1}$ captures indirect import exposure: sector i is affected not just by its direct import share ω_{Fi} , but also by its suppliers' import shares (through $\Omega \omega_F$), its suppliers' suppliers (through $\Omega^2 \omega_F$), and so on, each step discounted by price rigidities

along the chain.

4.3 Breakdown of divine coincidence

Condition for no cost-push	Closed economy	Open + float
No productivity cost-push: $\phi^T \mathcal{V} = \mathbf{0}^T$	Required	Required
No ER cost-push: $\phi^T \Gamma = 0$	n/a	Also required
Number of conditions on ϕ	1	2
Solution	Unique DC index	Generically none

The left null space of $\mathcal{V}$ is one-dimensional (spanned by the DC weights). The constraint $\phi^T \Gamma = 0$ is a hyperplane in $\mathbb{R}^N$. Generically these do not intersect, so **no single inflation index eliminates both cost-push channels under a float**.

Exception: if $\Gamma \propto \mathcal{B}$ (ER affects sectors in exact proportion to their labor intensity), divine coincidence survives. With real TiVA data: no.

4.4 Three FX regimes

	Float	Peg	Managed float
ER cost-push $\Gamma \Delta e_t$	Non-zero	Zero ($\Delta e = 0$)	Partially offset
MP independence	Full	None (trilemma)	Partial
DC index survives?	No (2 cost-push channels)	Yes (trivially)	Modified DC
Optimal target	No simple index	DC index (but no MP instrument)	Use both i_t and ν_t

Under **managed float**, the CB uses FX intervention ν_t as a second instrument. Modified UIP: $i_t - i_t^* = \mathbb{E}[\Delta e_{t+1}] + \psi_t + \nu_t$. The optimal intervention neutralizes the ER cost-push on DC inflation:

$$\nu_t^* = (\mathbf{w}^{DC})^T \Gamma \cdot \Delta e_t$$

This restores a **modified divine coincidence**: with ν_t^* , the CB can independently target the output gap (via i_t) and eliminate ER cost-push (via ν_t).

4.5 UIP and IS curve (open economy)

UIP:

$$i_t - i_t^* = \mathbb{E}_t[\Delta e_{t+1}] + \psi_t + \nu_t$$

IS curve (modified from closed economy):

$$\tilde{y}_t = \mathbb{E}_t \tilde{y}_{t+1} - \frac{1}{\gamma} (i_t - \beta^T \mathbb{E}_t \boldsymbol{\pi}_{t+1} - r_t^{\text{nat}}) - \frac{\alpha_F}{\gamma} \mathbb{E}_t [\Delta e_{t+1}]$$

where $\alpha_F = \beta^T \boldsymbol{\omega}_F$ is the import share in the CPI. The last term: expected ER depreciation $\Rightarrow$ expected rise in import costs $\Rightarrow$ fall in real consumption $\Rightarrow$ lower output gap.

5 Formula-to-Code Quick Reference

Formula	Matlab code
$L = (I - \Omega)^{-1}$	<code>L = inv(eye(n)-Omega)</code>
$\boldsymbol{\lambda}^T = \boldsymbol{\beta}^T (I - \Omega)^{-1}$	<code>lambda = beta'*inv(eye(n)-Omega)</code>
$\hat{\delta}_i = \delta_i(1 - \rho(1 - \delta_i))/(1 - \rho\delta_i(1 - \delta_i))$	<code>hat_d = delta.*(1-rho*(1-delta))./(1-rho*delta.*(1-delta))</code>
$\Delta = \text{diag}(\hat{\delta})$	<code>Delta = diag(hat_delta)</code>
$(I - \Omega\Delta)^{-1}$	<code>inv(eye(n)-Omega*Delta)</code>
$\tilde{A} = \Delta(I - \Omega\Delta)^{-1}$	<code>A_tilde = Delta*inv(eye(n)-Omega*Delta)</code>
$\kappa = 1 - \boldsymbol{\beta}^T \tilde{A} \boldsymbol{\alpha}$	<code>kappa = 1 - beta'*A_tilde*alpha</code>
$\mathcal{B} = (\gamma + \varphi) \tilde{A} \boldsymbol{\alpha} / \kappa$	<code>b = (gamma+phi)*A_tilde*alpha/kappa</code>
$\mathbf{v} = \tilde{A}(\boldsymbol{\alpha}(\boldsymbol{\lambda}^T - \boldsymbol{\beta}^T \tilde{A})/\kappa - I)$	<code>v = A_tilde*(alpha*(lambda-beta'*A_tilde)/kappa - eye(n))</code>
$\mathbf{w}^{DC} \propto \lambda_i(1 - \hat{\delta}_i)/\hat{\delta}_i$	<code>llambda = lambda*(eye(n)-Delta)*inv(Delta)</code>
$MM = I + \Delta \mathbf{v}(I - \Omega)$	<code>MM = eye(n) + Delta*v*(eye(n)-Omega)</code>
<i>SOE extension (new objects)</i>	
$\Gamma = \tilde{A} \boldsymbol{\omega}_F / \kappa$	<code>Gamma = A_tilde*omega_F/kappa</code>
$\alpha_F = \boldsymbol{\beta}^T \boldsymbol{\omega}_F$	<code>alpha_F = beta'*omega_F</code>
$(\mathbf{w}^{DC})^T \Gamma \neq 0 \Rightarrow \text{DC breaks down}$	<code>dc_Gamma = dc_weights*Gamma</code>

6 Closed vs. Open Economy: Summary Table

Object	Closed economy	Open economy
Marginal cost	$\alpha w + \Omega \mathbf{p} - \log \mathbf{A}$	$+\omega_F(p^* + e)$
Phillips curve	$\rho(I - \mathcal{V})\mathbb{E}\boldsymbol{\pi}' + \mathcal{B}(\gamma + \varphi)\tilde{y} - \mathcal{V}\boldsymbol{\chi}$	$+\Gamma\Delta e_t$
ER pass-through Γ	n/a	$\tilde{A}\boldsymbol{\omega}_F/\kappa$
IS curve	$\tilde{y} = \mathbb{E}\tilde{y}' - (i - \mathbb{E}\pi^C - r^{\text{nat}})/\gamma$	$-(\alpha_F/\gamma)\mathbb{E}[\Delta e']$
New equation	—	UIP: $i - i^* = \mathbb{E}[\Delta e'] + \psi + \nu$
Divine coincidence	Unique DC index (no cost-push)	Breaks down under float
Pass-through amplifier	—	$(I - \Omega\Delta)^{-1}$
Optimal FX policy	—	$\nu_t^* = (\mathbf{w}^{DC})^T \Gamma \cdot \Delta e_t$